09 — Feed Source Audit & Fetch Fix (2026-10-14)

Motivation. Production (Fly.io, Singapore) was silently fetching fewer news sources than configured. A live audit compared what `RSS_Sources.txt` promises against what actually returns news, both from the developer machine and from inside the Fly machine.

Audit method

1. Parsed `RSS_Sources.txt` (18 active URLs; 2 commented-out Telegram entries).
2. Fetched every URL from the local machine with a browser User-Agent (script kept at `scripts/check_feeds.py`).
3. Read the live Fly logs (`flyctl logs -a apanewsharitok`) for recurring prefetch errors.
4. Re-tested the four 403 feeds **from inside the Fly machine** (`flyctl ssh console`) with a real Chrome UA to separate UA-blocking from IP-blocking.

Results

#	Source	Local (browser UA)	Fly (bot UA)	Fly (browser UA)	Cause
1	sarawaktribune.com	[OK] 10 items	[OK]	—	—
2	news.seehua.com	[OK] 30 items	[X] 403	[X] 403	Datacenter-IP block
3	theborneopost.com	[OK] 20 items	[X] 403	[X] 403	Datacenter-IP block
4	malaysiakini.com	[OK] 10 items	[OK]	—	—
5	malaymail.com	[OK] 50 items	[OK]	—	—
6	bernama.com	[OK] 10 items	[OK]	—	—
7	lowyat.net	[OK] 10 items	[OK]	—	—
8	freemalaysiatoday.com	[OK] 50 items	[X] 403	[X] 403	Datacenter-IP block
9	hmetro.com.my/utama.xml	[OK] 30 items	[OK]	—	—
10	hmetro.com.my/mutakhir.xml	[OK] 20 items	[OK]	—	—
11	dayakdaily.com	[OK] 10 items	[OK]	—	—
12	rss.app feed	[X] 402 Payment Required	[X] 402	—	Lapsed paid subscription
13	sarawakedition.com	[OK] 10 items	[OK]	—	—
14	cms.buletintv3.my	[OK] 10 items	[OK]	—	—
15	berita.rtm.gov.my	[OK] 10 items	[OK]	—	—
16	news.google.com "Sarawak water"	[OK] 100 items	[OK]	—	—
17	news.google.com "Sarawak energy"	[OK] 100 items	[OK]	—	—
18	tvsarawak.my	[OK] 10 items	[X] 403	[OK] 200	Bot User-Agent block

Tally: 18 configured → 17 working locally → **13 working on Fly** before the fix.

Note: `seehua` and `buletintv3` return slightly malformed XML (leading whitespace / unbound prefix), but `feedparser` tolerates both — no code change needed.

Changes made (2026-10-14)

1. `src/scrapers/rss_reader.py` — the fetch User-Agent changed from the self-identifying `Mozilla/5.0 (ApaNewsHariTok bot)` to a real Chrome 129 UA plus an explicit `Accept: application/rss+xml, ...`

header. Evidence: with this UA, TV Sarawak returns 200 from the Fly machine (was 403 on every 20-minute prefetch cycle).

2. `RSS_Sources.txt` — the dead `https://rss.app/feeds/EYJxxljrFQtbWR8K.xml` line is commented out (402 Payment Required: paid subscription lapsed).

After redeploy, production should fetch **14 of 17 remaining sources**.

Known issue — remaining 3 sources (datacenter-IP blocked)

See Hua, Borneo Post, and Free Malaysia Today block Fly.io's IP range at the CDN/WAF level; no header change helps (verified from inside the Fly machine). Options, in order of preference for a future phase:

1. Route only these feeds through a feed-proxy (self-hosted RSSHub, or a small proxy worker) so the egress IP is residential/cloud-generic.
2. Replace with publisher-approved APIs where offered.
3. Accept the loss: these 3 remain available when the bot runs from a non-datacenter network (e.g. local dev).

Do not attempt UA rotation/cookies for these — the block is IP-based, not UA-based.

2026-10-14 additions (same day)

Source-expansion sweep: 25+ candidates tested locally, then re-verified **from inside the Fly machine** so production behavior is known in advance. Added to `RSS_Sources.txt`:

Source	Feed URL	Items/cycle	Rationale
Astro Awani	`https://www.astroawani.com/rss.xml`	10	Strong Sarawak coverage, national TV news
Berita Harian	`https://www.bharian.com.my/feed`	50	Major Malay daily; complements Harian Metro
The Sun Daily	`https://www.thesun.my/rss`	10	National English, low load

Verified-but-not-added (kept in reserve; The Vibes and Malaysian Insight return 100 items/cycle each, which is heavy for a 1 GB machine): `https://www.thevibes.com/rss`, `https://www.nst.com.my/feed`, `https://www.themalaysianinsight.com/feed/`, `https://focusmalaysia.my/feed/`, `https://www.therakyatpost.com/feed/`, `https://www.malaysianow.com/feed/`

No public RSS (rejected): Utusan Borneo (biggest Sarawak gap), Sin Chew, China Press, Oriental Daily, Sinar Harian, The Star, The Edge, official Sarawak gov / UKAS portals. These would need an RSSHub/proxy route (same solution as the 3 IP-blocked sources).

Infra: `fly.toml` VM memory raised 512 MB → 1 GB (shared-cpu-1x) to head off memory pressure from the wider source list; deployed 2026-10-14.

Reproduce

```
python scripts/check_feeds.py # local check of every source
flyctl logs -a apanewsharitok # live prefetch errors
flyctl ssh console -a apanewsharitok -C "python -c \"...\"" # fetch from Fly's network
```

Deployment note: requires `fly deploy` for both changes to reach production (`RSS_Sources.txt` is baked into the image via `COPY . .`).